

Verification

– important to verify that target protein is not lost during each stage of protein purification, or to check if the protein lost its activity or not

The verification occur while we are still doing purification, we will be tracking the protein to make sure that the protein is not lost and the activity of protein is not lost also

What we will do is to collect to samples to check your protein each step, for example after you lyse your cell you will take sample to make sure that there is protein, after you do precipitation you will take another sample, after dialysis also you will take sample, after size exclusion you will take sample, after ion exchange you take sample, after hydrophobic interaction you take sample

The sample is small quantity of protein that you take before and after each purification step, and as we said before we do all of this to know where is your protein, how much of your protein you still have, is your protein still active or lost its activity, so you are making sure that your protein is still there and you are making sure you are doing purification

After you collect these sample, you will do verification

of course, you have to do verification at the end, to make sure that you have purified your protein, what is the concentration of your protein, is your protein still active or not

one technique for verification is SDS page

- SDS PAGE (polyacrylamide gel electrophoresis)

Page stand for (polyacrylamide gel electrophoresis)

This a gel electrophoresis for protein but instead of using agarose we use a type of gel called polyacrylamide gel

So basically, we are separating protein based on their size, and using an electrical current

- Detergent called SDS (sodium dodecyl sulfate) is added to protein mixture and mixture is heated

- Sulfate charges are evenly distributed along denatured protein making separation dependent on size and not charge of protein

To separate the protein based on their size we have to make all the proteins have the same charge by using the SDS

- After treatment with SDS protein sample is loaded onto PAGE where it forms a single band at a specific location on the gel depending on its molecular size and mass

So, let's say we have three proteins before separating them on the gel, these proteins will be treated with SDS, now the SDS will do two things:

- 1- It will denature these proteins
- 2- And it will give them a negative charge

After that they will be ready to be separated using gel electrophoresis, now what we will get on the gel is 3 band (assuming each protein consist of one polypeptide), the separation is just like DNA the smallest will migrate faster

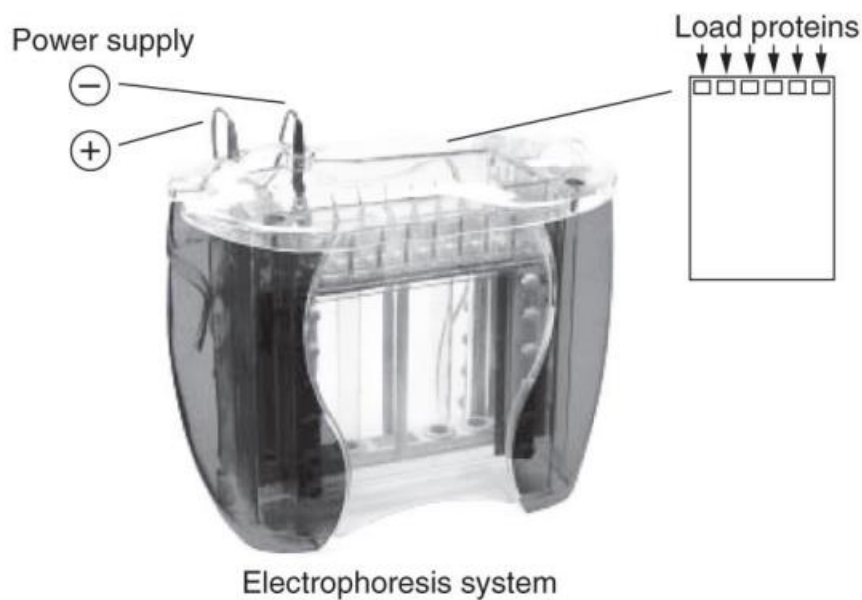
Sometimes you will have proteins that is made from more than one subunit

So basically, SDS separate polypeptide from each other and it can denature these polypeptide

– Coomassie stain - dye that combines with proteins so a colored band results

This stain binds to protein giving them color, typically this color will be dark blue

The protein is colorless, the gel is also colorless, so you have to add the stain to the gel so the protein bands will appear with dark blue color



This is the electrophoresis apparatus that is used to separate protein

The electrophoresis that is used to separate DNA is horizontal, this device is vertical, which means that protein moves from top to the bottom with the help of electricity

It's the same as the electrophoresis of DNA but here it's vertical not horizontal

The protein sample should be mixed with sample buffer

Sample buffer: glycerol + SDS + loading dye (tracking dye) + B-mercaptoethanol

Glycerol: will increase the density of your protein sample so when you add your protein to the well it will not go outside the well

SDS: denature the protein and give it negative charge

Loading dye: to know when to stop the electrophoresis

B-mercaptoethanol: it's a chemical used to break the disulfide bonds in protein (the ss bonds)

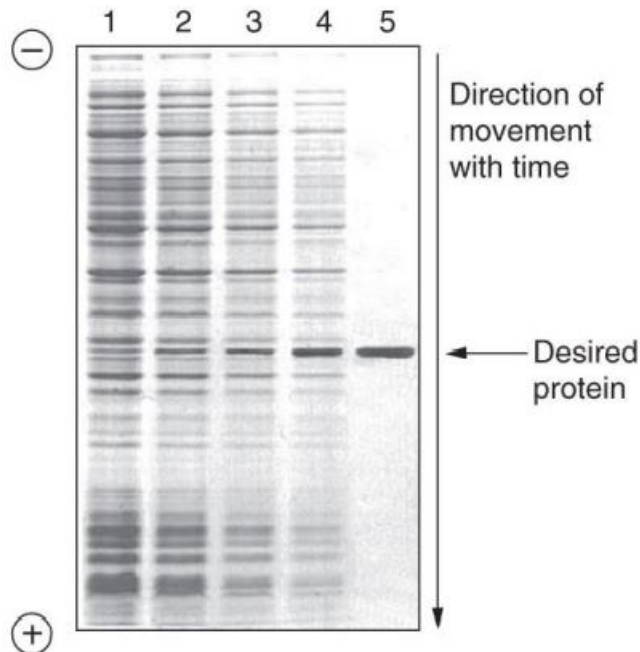
The ss bonds are the only covalent bond holding the 3d structure of the protein and the B-mercaptoethanol can disrupts them

We have cysteine amino acids, and these amino acids have thiol group that can form ss bonds

Then after you mix your protein sample with the sample buffer you can load your protein into the gel, run the gel, when the tracking dye reach the end of the gel you stop the electricity and then you take the gel and stain it with Coomassie stain to see the bands

» Compare stained sample with known size marker to estimate size of protein

» During purification process the colored band representing the protein should become increasingly intense showing that the protein is not lost during the procedure



Each band may represent a whole protein if it consists of one polypeptide, or a part of the protein (one polypeptide)

Now as we can see in the pic we have different lane and, each lane represent like we say in the beginning different purification step, now the purification is basically removing as much as we can of the contaminant protein to get in the end a solution the contain the desired protein, the first sample you take will contain the highest amount of contaminant protein and the last should contain the least amount of the contaminant otherwise you are wasting time and money, so If your protein consist of one polypeptide you should get one band that and the size of this band should correspond to the size of your protein

Here we will use protein marker just like the in the DNA electrophoresis, its job is to give us an idea about the size of the band, the protein marker is a solution of multiple protein with known sizes so when you run the marker you will get multiple band but you know the size of each band, so it will be just like a ruler

Now let's say that after the end SDS PAGE we end with one band does that mean that this band is 100% my protein?

The answer is no because there could be another protein that have the same size as my protein so to be sure about this protein we have to use the western blotting

- Specific method for proteins separated by SDS/PAGE= Western blotting

- Procedure

- : a. Proteins are transferred from the SDS/PAGE to a nitrocellulose membrane using electrical current

- b. All sites on membrane are blocked so that the antibody will not bind to them nonspecifically

- c. Primary antibody is added and incubated with the membrane

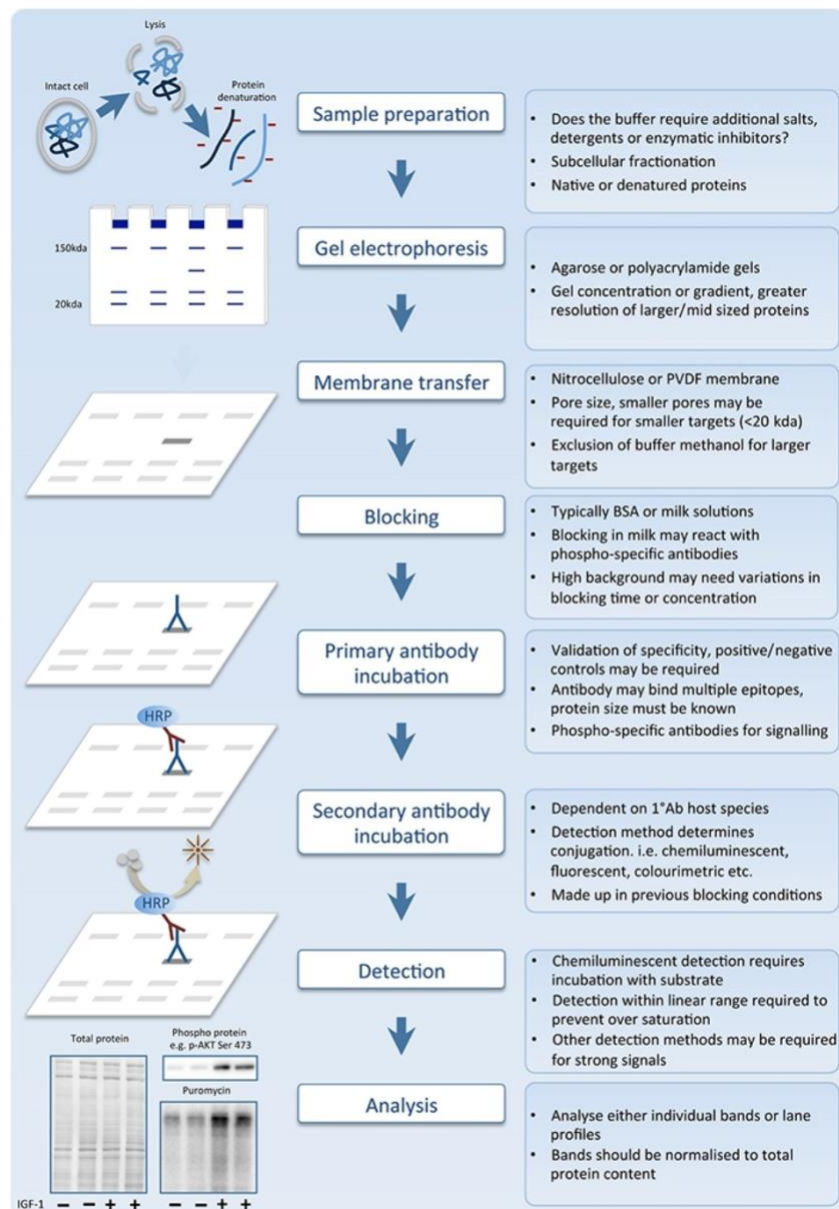
- If there are any antibodies present that are directed against one or more of the blotted antigens (proteins on the membrane) those antibodies will bind to the protein and the other antibodies will be washed away at end of incubation

- d. To detect antibodies that have bound to protein a secondary anti-immunoglobulin antibody coupled to a reporter group (alkaline phosphatase) are added to the membrane
- e. Excess secondary antibody is washed off the membrane
- f. Substrate is added that precipitates upon reaction with conjugate resulting in visible band where primary antibody is bound to the protein

western protein: separation of protein and detection of specific type of protein using anti bodies

in western blotting you take the SDS PAGE without staining and you do blotting, so after we separate the protein on the gel in blotting we are going to transfer the protein from the gel into the membrane(nylon membrane, cellulose membrane) we will also use the electricity to move the protein from the gel into the membrane , now we will continue working with the membrane, the next step (the first step in western blotting) is to do blocking, this means the surface of the membrane will be covered with protein because all the surface should be covered with protein so we will use a protein called BSA (bovine serum albumin) we can take this protein from blood cows, or we can use milk protein, so now the surface of membrane of the membrane is covered by proteins some of them are contaminant protein, some are my protein, and the rest is BSA, the next step would be to add primary antibody now this antibody should be specific to your protein (bind only to your protein), the next step would be to add secondary antibody (enzyme conjugated antibody) now this is a secondary antibody that will bind to the primary antibody not to your protein, we have an enzyme attached to the secondary antibody this enzyme could be AP (alkaline phosphatase) HRP (horseradish peroxidase), now you

will add a substrate(colorless) and the enzyme will convert the substrate into colored product which will color your protein, so in the



end you will have a piece of paper with one band if your protein consist of one polypeptide

Enzyme linked immunosorbent assay (ELISA)

Require two antibodies, one to capture the unique proteins and one attach to an enzyme to produce a color reaction.

ELISA occur on a multiwell plate using affinity chromatography.

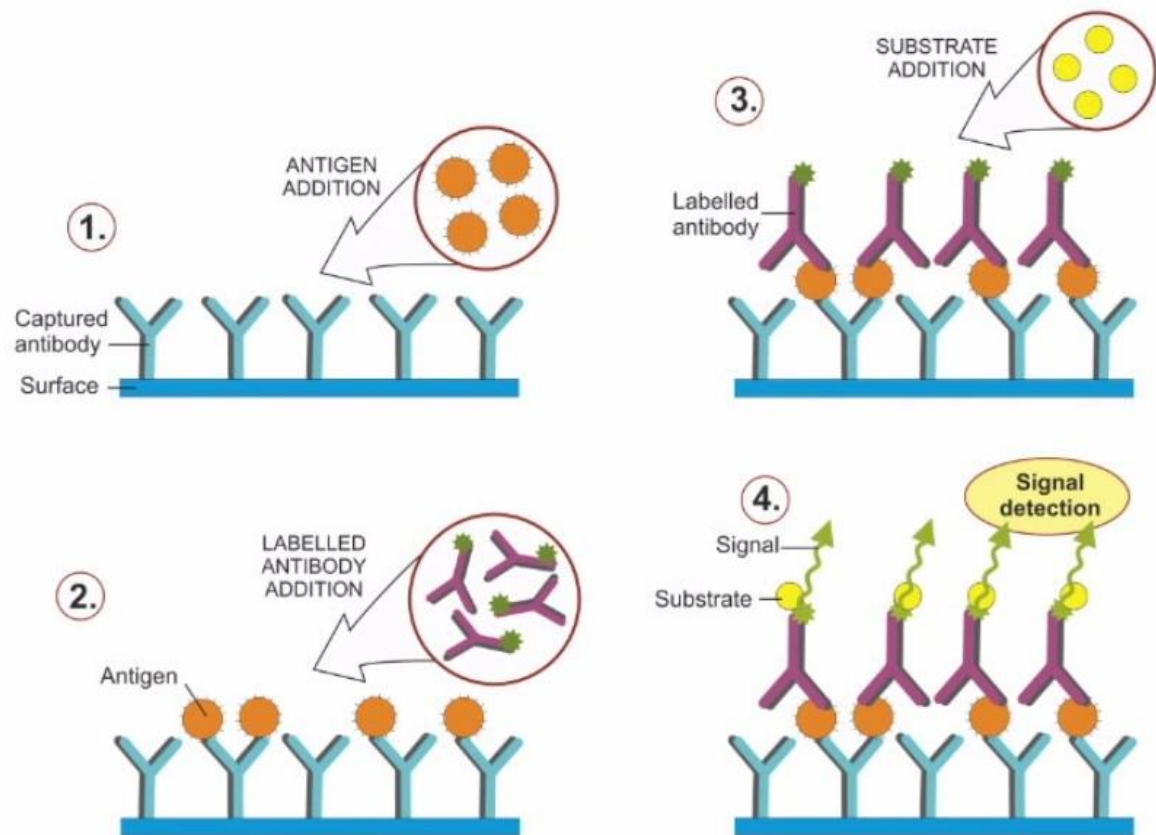
The first antibody to unique protein is plated on to multiwell ELISA plate, then the protein is added, after of series of washing and blocking steps – the second antibody (enzyme attached) is added.

The addition of substrate permits a color reaction to occur if the Antibody has bond.

Antibody 1 = attach to protein.

Antibody 2 = attached to an enzyme.

ELISA plates are designed to capture many important proteins currently under research.



Preserving proteins

After target protein has been isolated, collected and purified, it must be saved in a manner that will preserve its activity until can be used.

Lyophilization (freeze drying)

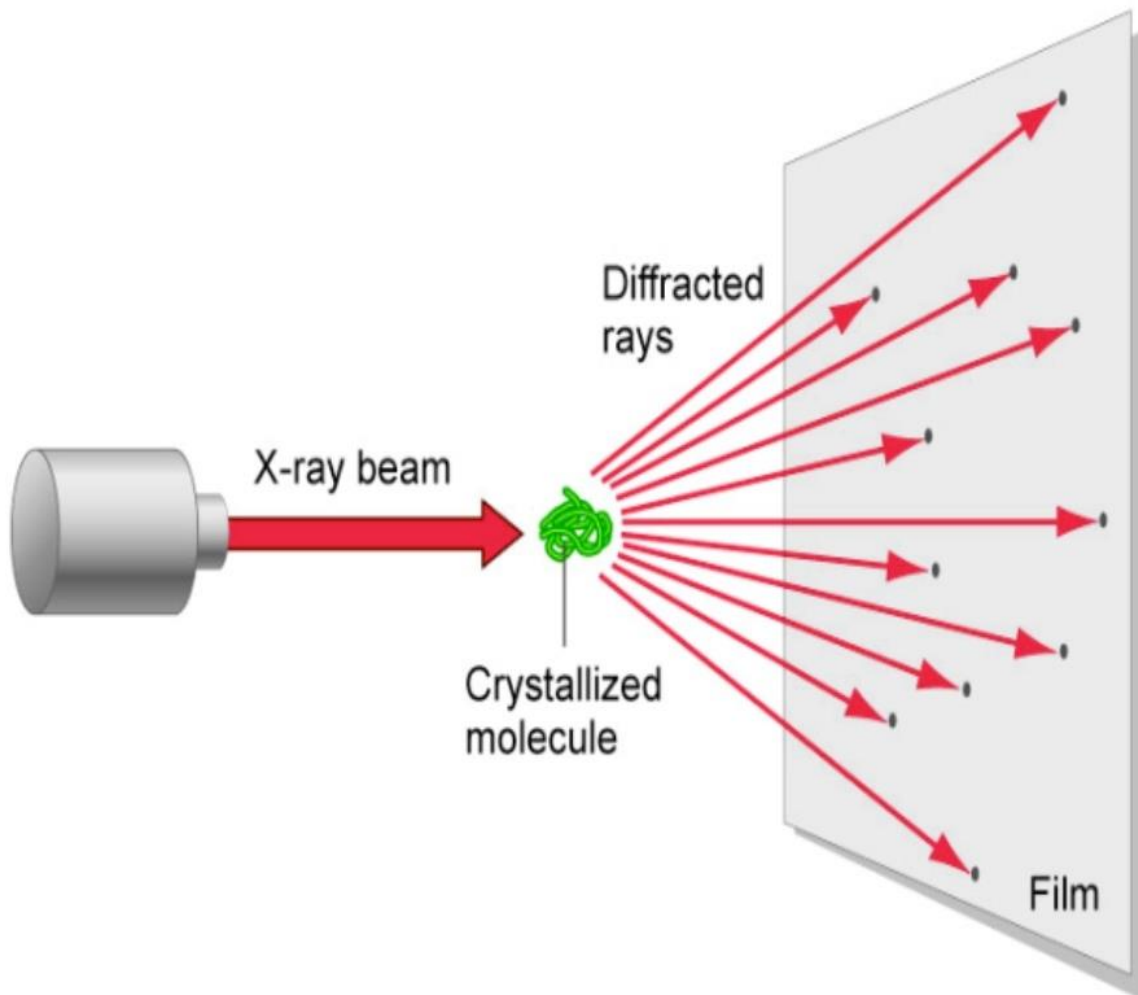
Purified liquid protein solution is frozen.

Ice crystals are converted directly to water vapor without melting into liquid water.

Lyophilization maintains protein structure

many freeze – dried proteins may be stored at room temperature for long periods of time.

X – ray crystallography used to determine the complex tertiary and quaternary structures of proteins.



First we need a crystal of protein, by using X-ray beams source, X rays will release toward a protein crystal, when the X rays collided with crystal, X rays will scatter, spots and dots will form, based on the spots and dots we can determine the 3D structure for specific protein.

Proteomics

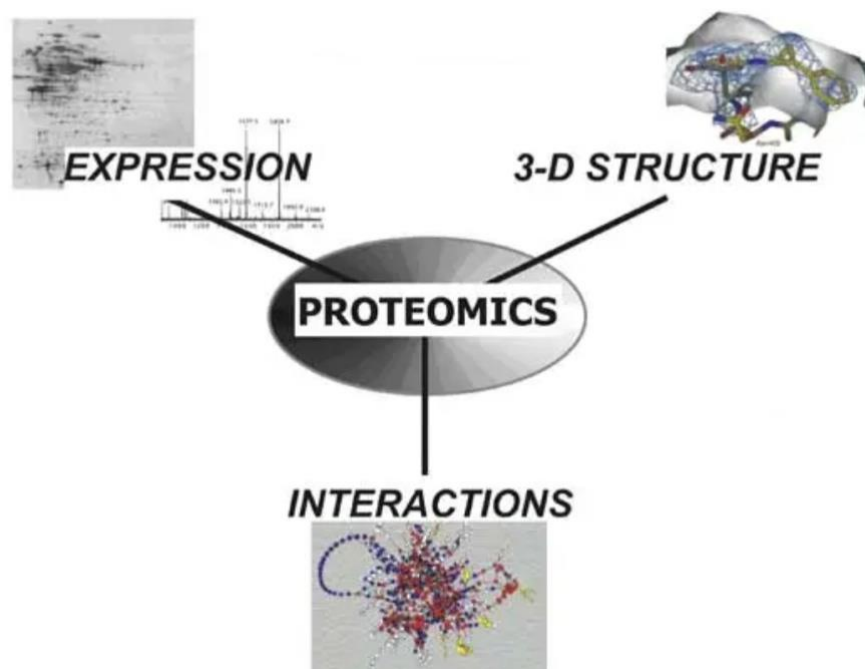
Is Dedicated to understanding the complex relationship of Disease and protein expression.

Proteomes : are compared between healthy and Disease States.

The variation of protein expression of specific are then correlated to onset or progression of specific Disease.

The goal of Proteomic research is the discovery of proteins markers that can be used in new diagnostic methods.

Proteomic also defined, study of all proteins in the all cells inside the body.

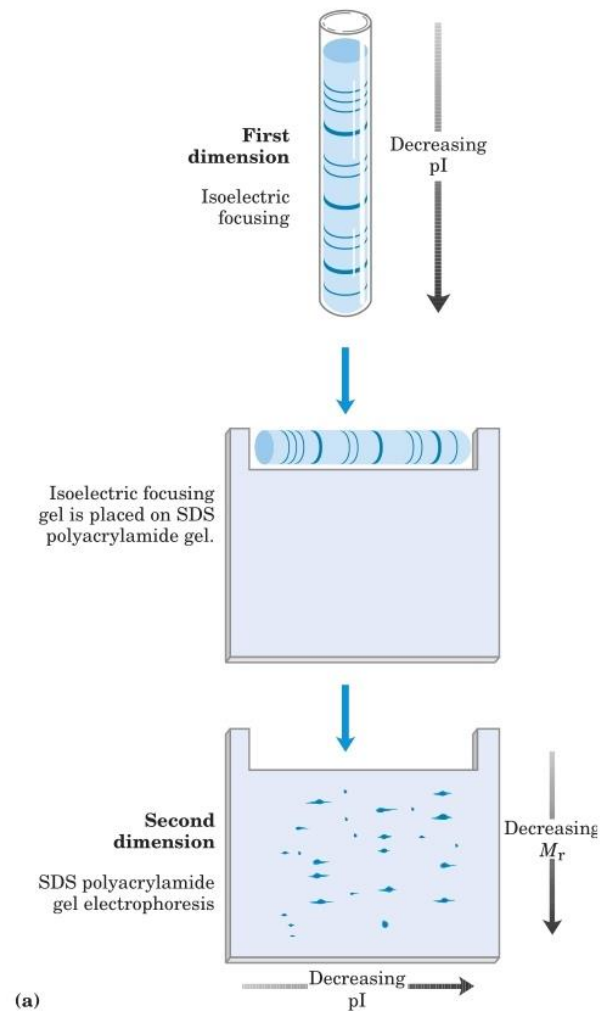


Techniques used for Proteomics

Two – dimensional electrophoresis

Is used to separate proteins, followed by mass spec.

Use amino acid sequencing to Characterized the protein.



Protein Microarray

Is a set of proteins immobilized on the surface – usually a glass slide – which has been coated with a Reagent will indicate the binding by color changing.

Functional protein micro array have Recently been Applied to discovery proteins interaction, protein – lipid, protein – DNA, protein drug, and protein-protein interaction.

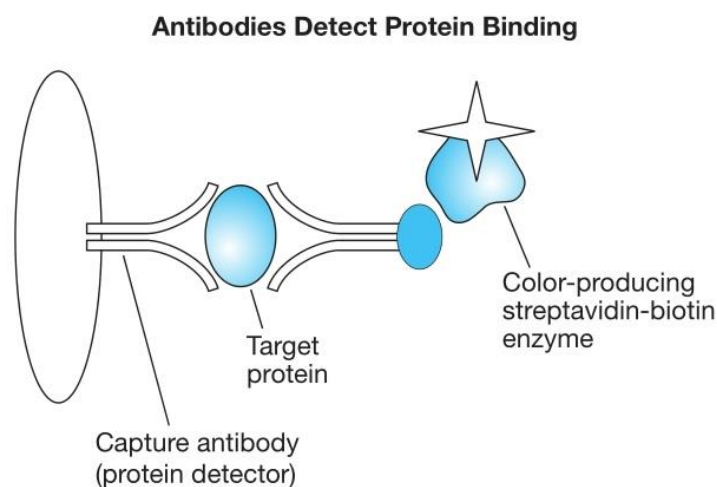


FIGURE 4.16 Protein Microarrays Protein microarrays that uniquely bind to single proteins are being perfected. When a protein binds, it releases a signal as the enzyme bound to the interacting protein degrades a substrate. Enzyme-linked immune sorbent assays (ELISA) bind protein targets in this way for detection.

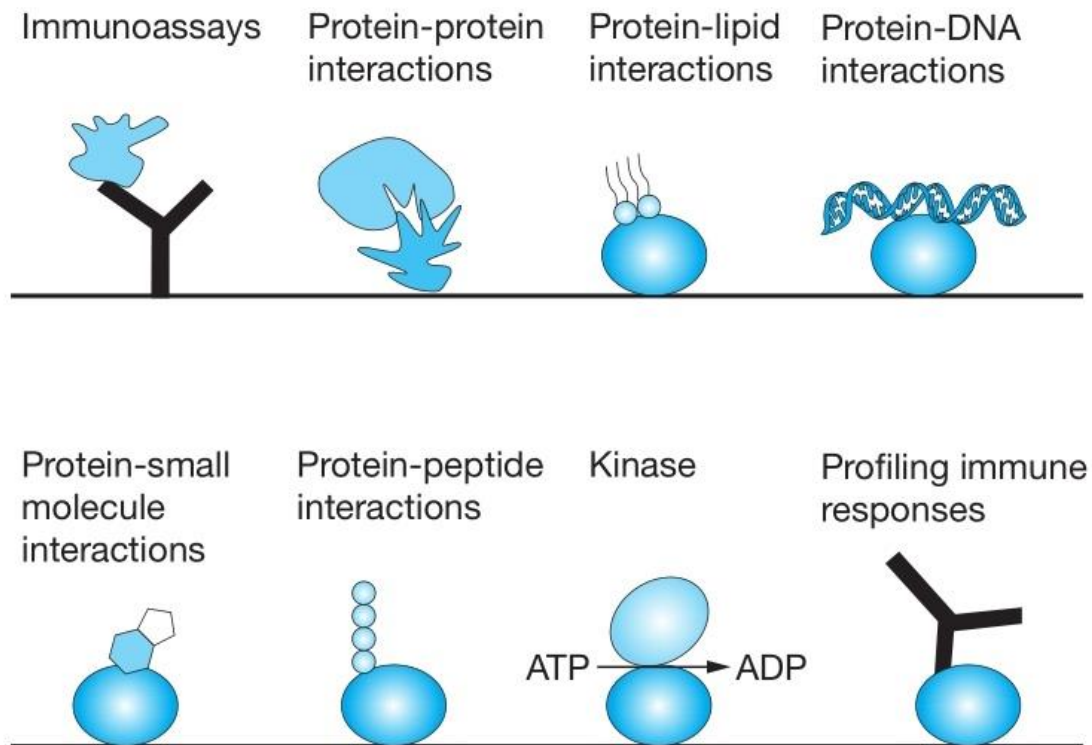


FIGURE 4.17 Protein Microarrays Can Detect More Than Just Proteins The capacity of protein microarrays has increased. Functional protein microarrays have recently been applied to the discovery of protein interactions (i.e., protein-protein, protein-lipid, protein-DNA, and protein-drug interactions), expanding knowledge of the significance of protein interactions to normal cell function. Binding antibodies or other proteins to the matrix allows detection when the binding allows a colored product to be produced.

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